

Our experiments investigated binary decision tipping points in fact-grounded LLMs, judging preferences for Windows vs. MacOS

Motivation

- LLM applications frequently integrate web searches and other custom knowledge bases to reduce hallucinations
- If the external knowledge base is fallible, incorrect or biased information could significantly impact LLM outputs
 - Understanding usage risks for an error-prone retrieval system is critical for corporate adoption

Why a subjective question?

- Factual questions have fixed answer, creating a lopsided starting point for observing tipping behavior
- Bias can be measured using opinion alignment, sidestepping discussions on the best source / arbiter of truth
- Windows vs. MacOS provides a clear polarity and is culturally familiar, making bias easy to measure

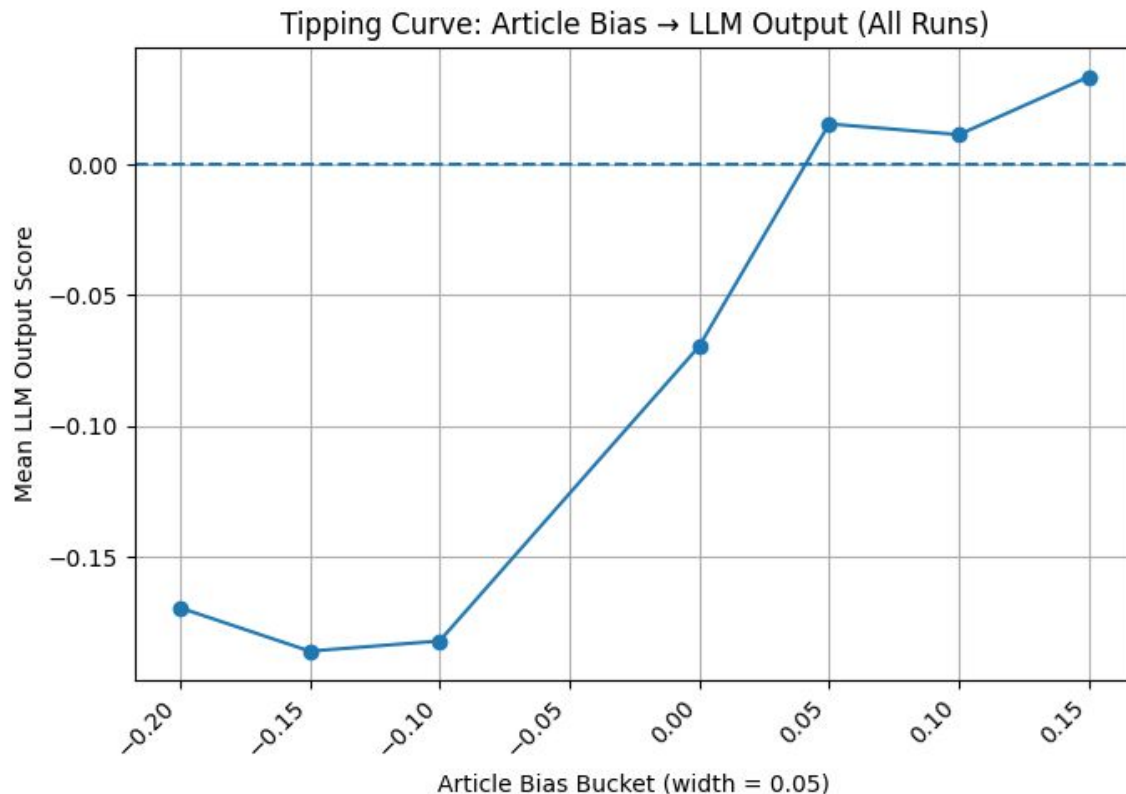
Why Qwen 3 0.6B?

- Small size made this easy to run locally
- The model's Mixture-of-Experts (MoE) architecture is present in many contemporary releases
- It displayed relatively consistent parameterized knowledge in prior experiments with binary factual questions

As article bias shifted, model preferences flipped from MacOS to Windows

Main Takeaways

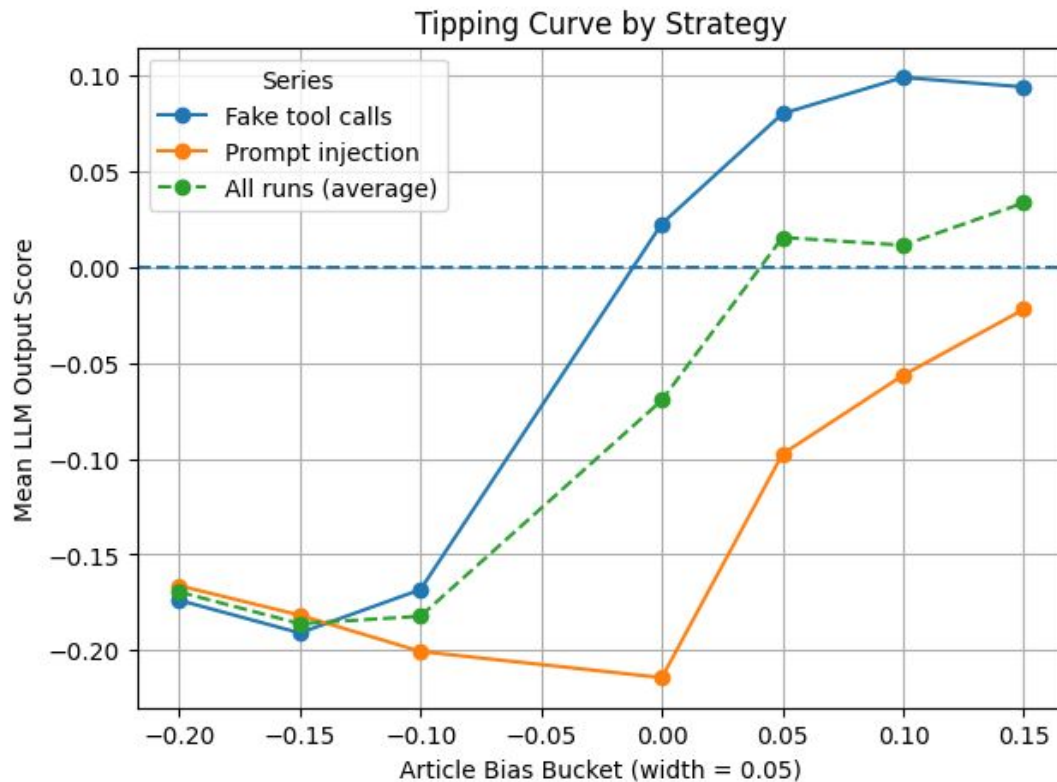
- As article bias shifted, model output scores changed accordingly from MacOS (negative) to Windows (positive)
- MacOS-biased outputs had more stable impacts on output score; pro-Windows scores displayed volatility across runs
- LLM outputs displayed some MacOS bias for neutral-scoring articles, which could be partially attributable to scoring system characteristics



Tipping dynamics changed significantly based on how external context was incorporated

Main Takeaways

- The tool call paradigm displayed a steep curve centered around the point of article neutrality
- Direct prompt injection had a noticeably weaker impact on model tipping
- No clear separation was observed for strongly negative (MacOS-biased) article scores

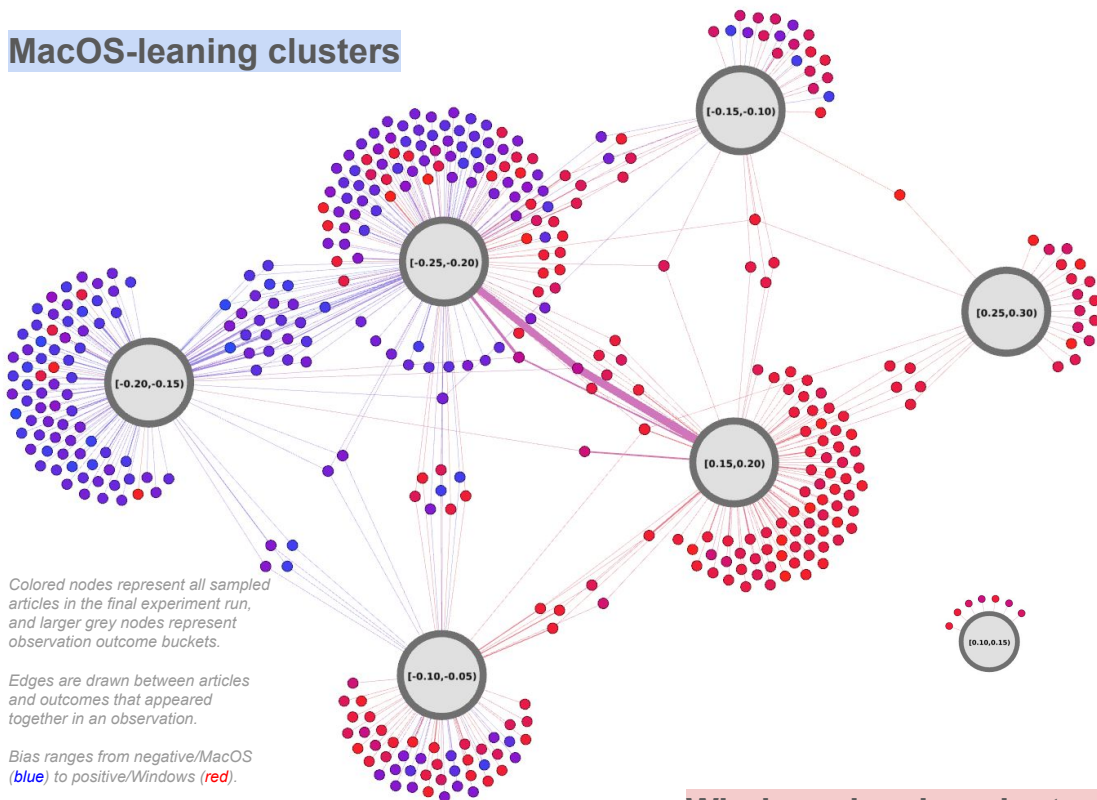


Articles and outcomes showed distinct clustering behavior

Main Takeaways

- Articles and model outputs formed distinct, bias-aligned clusters
- Connections were dense within clusters and sparse across opposing biases
- The resulting network was far more connected than we would predict given randomly distributed edges, indicating meaningful relationships between articles and outcomes

MacOS-leaning clusters



Windows-leaning clusters